

5. Planet **J** is similar in size to the Earth. However, the temperature on planet **J** is about 470°C and the clouds in its atmosphere are made of sulfuric acid.

A group of students investigated the properties of some metals. Their aim was to see if they could find a metal suitable for designing a spacecraft to explore planet **J**. Their findings are shown below.

Zinc – fizzes quite vigorously with sulfuric acid, has a melting point of 420°C and a density of 7.1 g/cm^3

Copper – has a melting point of 1083°C , does not react with sulfuric acid and has a density of 8.9 g/cm^3

Sodium – has a density of 1.0 g/cm^3 , reacts explosively with sulfuric acid and has a melting point of 98°C

Magnesium – has a density of 1.7 g/cm^3 , a melting point of 650°C and it fizzes vigorously with sulfuric acid

Titanium – does not react with sulfuric acid, has a melting point of 1675°C and a density of 4.5 g/cm^3

Lead – has a melting point of 328°C , a density of 11.3 g/cm^3 and does not react with sulfuric acid

The spacecraft needs to withstand the conditions on the surface of planet **J**. The mass of the spacecraft also needs to be as low as possible in order for it to have enough energy to escape the Earth's gravity.

- (a) Which **one** of these statements best describes why magnesium is an **unsuitable** metal for the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

its density is 1.7 g/cm^3

☐

its melting point is 650°C

☐

it fizzes vigorously with sulfuric acid

☐

it is malleable

☐

- (b) Small amounts of lead are sometimes used in electrical circuits.

Which **one** of these statements best describes why lead would **not** be suitable for use in the electrical circuits of the spacecraft? Put a tick (✓) in the box next to the correct answer. [1]

it does not react with sulfuric acid

☐

it is ductile

☐

it would melt when it lands on planet J

☐

its density is 11.3g/cm³

☐

- (c) The students decided that titanium is the most suitable metal from which to build the spacecraft.

Put a tick (✓) in the boxes next to the **two** statements that best describe the reasons for their choice. [2]

it does not react with sulfuric acid

☐

it is expensive

☐

it is a good conductor of heat

☐

it is non-magnetic

☐

it has a melting point much higher than the temperature on planet J

☐

it is shiny so will reflect the sun's rays

☐

- (d) Sodium reacts explosively with sulfuric acid. Sodium sulfate and hydrogen are produced. Complete and balance the equation for this reaction. [2]

